Supplemental Material for "Representation-Dependent Recoverability in Quantum Compilation"

Jinze Yang[1], Yangyang Li[2], and Xiu-Hao Deng[3]

[1]School of Physics, Xidian University, Xi'an 710071, China
[2]School of Artificial Intelligence, Xidian University, Xi'an 710071, China
[3]Shenzhen International Quantum Academy, Shenzhen 518048, China

1 Scope and evidence policy

This document contains expanded experimental protocols, negative results, status accounting, a historical/stress-result registry, and data provenance for the full-length main manuscript. The headline proof and the earlier application appendices remain in the active main manuscript so that their assumptions, constructions, and scope are visible together. Historical files are retained for auditability but are not promoted to canonical W3–W9 evidence unless they appear in `data/manifest.yaml`.

Only `completed_valid` or a campaign-specific `completed` paired with a passing independent certificate is an accepted quality outcome. `timeout`, `memory_error`, `unsupported`, `predicate_error`, `correctness_failure`, and `numerical_inconclusive` remain separate categories. A failed or unsupported cell is never assigned an infinite, zero, or best-available metric.

Table 1: Canonical campaign inventory. Counts are generated from registered frozen sources; status-only rows remain in their source tables.

Package	Canonical cells	Audit status
W3 tradeoff traces	592	all theorem diagnostics pass
W4 matched representations	1,080	1,065 completed; 15 predicate errors
W5 external baselines	1,728	1,695 completed; 33 predicate errors
W6 certificate audit	56 mutations	0 false accepts
W7 fixed-error QRE	288	all completed-valid
W8 natural factorial	21,060	all completed-valid
W9 normalized index	24,748	unique complete keys

2 Instrumented commitment–state–IR campaign

The W3 reference suite is a collection of formal-model objects, not wrappers around production compilers. The event schema records input position, pass, control/store/window bytes, parameter bytes, serialized live-IR bytes, committed output delta and total, RSS, and wall time. The theorem coordinate S is computed from the complete serialized cut state. RSS is an ancillary operational measurement and is never used as a proof surrogate.

The grid crosses m, update rounds, K, separation margin, forward passes, memory cap, schedule, and seed. Every run has a per-run manifest and event log. The aggregate file `data/frozen/tradeoff_summary.csv` reports both the exact transcript residual and the state-cap envelope residual. The frozen anomaly table is empty; this is reported as an observed campaign fact, not substituted for the theorem.

Table 2: Reference compiler behaviors. Memory percentages refer to the payload portion of the full aggregate vector; framing remains charged.

Compiler	Aggregate memory	Commitment policy
Echo	none	append each update in stream order
Full aggregation	full	retain vector, emit canonical aggregate
Hybrid-25	25%	aggregate public block, pass through remainder
Hybrid-50	50%	aggregate public block, pass through remainder
Hybrid-75	75%	aggregate public block, pass through remainder
IR materializing	full plus IR	retain explicit materialized stream

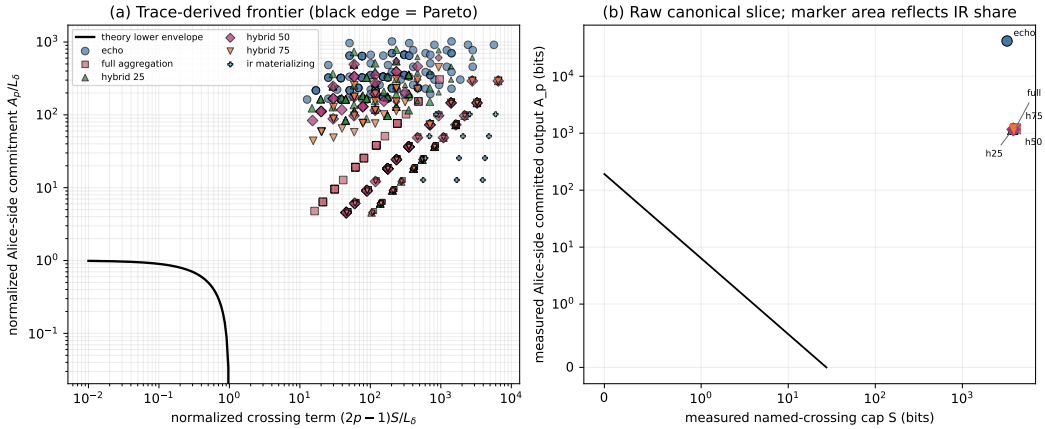


Figure 1: Complete event-traced W3 surface.

3 Matched representations and external semantic baselines

For each target the generator creates semantic aggregate, contiguous flat, round-robin flat, random commuting order, masked-share/update stream, locally folded flat, QASM bridge, TKET bridge, and ZX bridge forms. The representation record freezes its identifier, seed, generation parameters, serialized sizes, input certificate, and content hashes. Compilers see the same target basis, hardware abstraction, timeout, memory cap, and certificate.

The five W4 compilers are echo/pass-through, full aggregation, Qiskit optimization level 3, TKET PauliSimp, and PyZX full-reduce. W5 then applies eight frozen configurations: native and unified forms of the authors reference, Qiskit level-three local optimization, TKET PauliSimp, and PyZX full-reduce. Native-to-unified bridge deltas are measured before certificate comparison. A semantic-capable external output is grouped with the semantic side even when it is more compact than the authors' output.

The fifteen W4 predicate errors and thirty-three W5 predicate errors are retained in the canonical parquet files. They arise when an output lies outside the declared certificate predicate; they are neither correctness failures nor wins. No completed output bypasses the independent certificate.

4 Correctness certificate adversarial audit

The exact checker accepts commuting Pauli rotations in a tracked Clifford frame with rational-π and declared formal-symbol coefficients. It moves a Pauli sign into the coefficient, reduces the rational-π component modulo 2π, discards global phase, merges equal supports, and sorts addressed records deterministically. Equality of the canonical frame and rotation records implies projective unitary equality in the stated domain.

The mutation suite changes one angle, sign, support bit, qubit permutation, Clifford frame, deleted gate, or duplicated gate. Dense matrices through six qubits form an independent cross-check. The frozen result contains 56 mutations, 12 accepted and rejected cross-check cases, 0

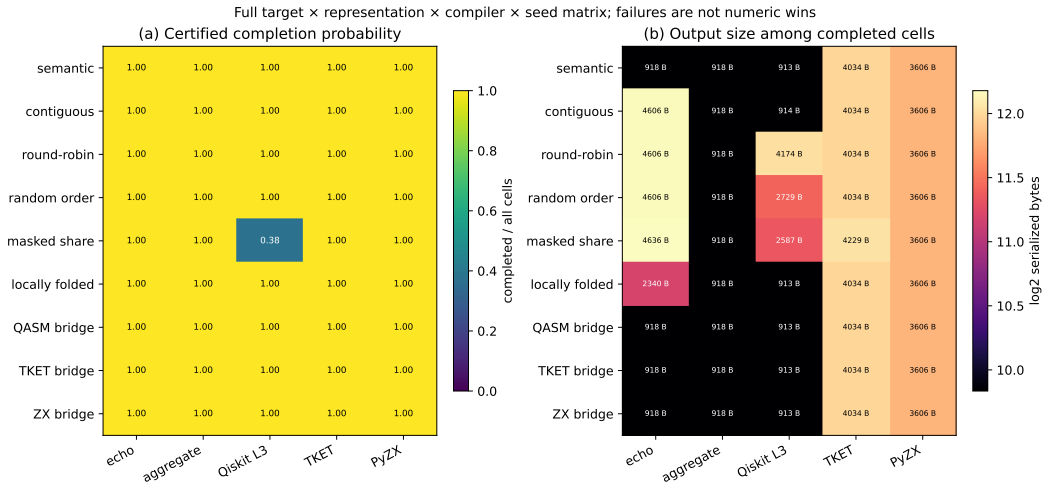


Figure 2: Full representation–compiler interaction summary; completion probability is separated from completed-cell metrics.

false accepts, and no dense/symbolic disagreement. Floating-only parameters, unlisted gates, and noncommuting extracted networks are explicitly outside the acceptance domain.

5 Fixed-total-error QRE details

The budget schema requires

$$\epsilon_{\text{alg}} + \epsilon_{\text{synth}} + \epsilon_{\text{logical}} \leq \epsilon_{\text{total}}. \tag{1}$$

The campaign uses the same fractional split in every pipeline and reallocates per-rotation synthesis accuracy after compilation. Equal-decimal allocation and a cost-optimized discrete allocation are both retained. The deterministic staq grid-synthesis executable runs with checking enabled; an angle cache is keyed by the exact angle/precision pair and records every use.

Table 3: QRE factorial controls.

Dimension	Frozen levels
Total error	$10^{-3}, 10^{-6}, 10^{-9}$
Allocation	equal decimal; cost optimized
QEC assumptions	conservative; improved surface-code profiles
Factory parallelism	serialized, low, medium, high Pareto points
Pipelines	semantic-first; materialize-first; external TKET PauliSimp
Outputs	rotations, T, qubits, factories, cycles, runtime, spacetime

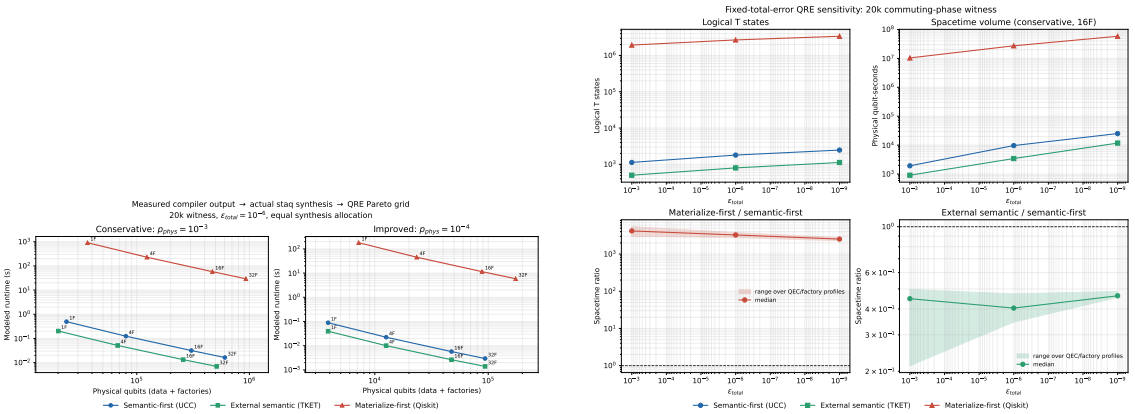


Figure 3: Pareto and sensitivity views generated from the complete W7 parquet file. Reversal or disappearance regions would remain visible; none is removed by filtering.

6 Natural workloads and complete ablation

The natural campaign is a full factorial over the declared families, scales, seeds, density levels, topologies, and all six binary mechanism toggles. Each cell records gates, depth, CX, runtime, RSS, input/output IR bytes, description bits, dense certificate, and the fixed-error QRE decomposition including data and factory physical qubits.

Table 4: Matched structured-family result retained without cherry-picking.

Family	External-minus-authors gates [95% interval]	Strict success
controlled_power_ladder	9.0 [5.7, 12.7]	yes
diagonal_hamiltonian	38.0 [25.9, 51.8]	yes
phase_polynomial_blocks	-24.3 [-33.8, -16.3]	no
qaoa_ising	44.7 [33.7, 56.6]	yes
qft_arithmetic	8.1 [1.9, 13.8]	no
qpe_controlled_powers	-12.0 [-22.0, -3.0]	yes
trotter_commuting_blocks	33.0 [17.6, 48.2]	yes

Table 5: All six natural-family gate-count main effects. Off-minus-on is positive when enabling the factor reduces gates.

Factor	Mean reduction	Bootstrap 95% interval
semantic_lift	72.2	[66.3, 78.1]
aggregation	80.4	[74.6, 86.4]
selector	0.0	[0.0, 0.0]
cache_reuse	0.0	[0.0, 0.0]
preset_recognizer	364.9	[349.8, 380.5]
projected_block_selection	0.0	[0.0, 0.0]

The preset recognizer is the largest isolated gate-count contribution. Semantic lift and aggregation also have positive mean effects. Selector, cache/reuse, and projected block selection have zero isolated gate effect in this reference suite, and this null result is retained. Both negative controls have zero all-on-minus-all-off differences for gates, depth, CX, logical T, and spacetime, with no systematic-regression interval.

The line-topology preflight caught an invalid final-layout permutation in a candidate router, whose dense certificate failed before the frozen campaign. We replaced it with an explicit adjacent CX-SWAP construction restoring identity layout and reran the entire campaign. The failed preflight is retained because correctness takes precedence over favorable metrics.

7 Legacy and stress-result registry

The active submission body is the full-length `PaperDraft/main.tex`. The unintegrated 38-page source is frozen as `PaperDraft/legacy_main_v7_2.tex`, and the superseded nine-page reconstruction is frozen as `PaperDraft/short_main_w10.tex`. Neither frozen file is the submission source. The active manuscript retains useful earlier exact-semantics and stress diagnostics, but labels their provenance and does not silently mix them with the W3–W9 canonical headline tables.

Table 6: Disposition of legacy material.

Material	Location	Submission disposition
Exact-semantics four-qubit witnesses	`appendix/appendix_C_prxq.tex`	Supplementary context only; not evidence for the K-ary asymptotic claim.
Finite-range rational CP period	`research/rational_witness/`	Retained certificate for the legacy exact-semantics corollary; not a W8 natural result.
Packing-width stress run	`research/packing_scaling/`	Historical theorem-linked stress evidence; superseded for headline measurements by W3/W4.
Width/topology external stress	`appendix/appendix_C_prxq.tex`	Status table retained; unsupported/predicate errors are not converted to quality.
Historical cut tracing	`research/cut_trace/`	Noncanonical after the event-schema revision; W3 event logs are canonical.
Earlier matched/natural tables	`research/matched_representation/`, `research/natural_factorial/`	Archived; W4 and W8 root-level parquet freezes are canonical.
Earlier per-rotation proxy QRE	`research/fixed_budget_qre/`	Excluded from the main claim; W7 fixed-total-error QRE supersedes it.

This separation is deliberate. It preserves failed and superseded evidence without allowing duplicate filenames or stale values to support a new claim.

8 Data lineage and independent-directory reproduction

The normalized record key is

$$(\texttt{experiment_id}, \texttt{instance_id}, \texttt{representation_id}, \texttt{method_id}, \texttt{seed}, \texttt{backend_hash}, \texttt{tool_version}, \texttt{artifact_commit}, \texttt{config_hash}).$$

The index contains 24,748 unique records and retains 48 predicate-error rows. CSV/parquet mirror checks, duplicate-label detection, receipt source hashes, row counts, and orphaned frozen artifacts are checked by `scripts/audit_results.py`.

The clean-room runner copies a minimal artifact to an independent temporary directory, creates a new venv bound to the locked project packages, and works only from that directory. It reruns 9 headline-theorem tests, independently compiles and densely certifies one matched QAOA/Ising instance with both reference pipelines, and runs a fixed-error QRE subset that exercises 17 deterministic synthesis pairs. This is working- directory isolation, not a claim of binary independence from the host Python installation.

The committed repository is sufficient for this frozen-data reconstruction. The approximately 700-MB raw W3–W8 run tree is versioned separately in the private `OIerYangJZ/paper1-data` archive at the revision recorded in `data/manifest.yaml`; the audit verifies its registered hashes when that optional archive is materialized. The one-command rebuild does not claim to rerun the complete measurement campaigns.

Every W3–W9 headline experimental number in the main text and this supplement is a generated macro registered in `data/provenance_map.yaml`. Retained historical diagnostics keep

their original source paths and cannot support a new headline number unless registered. The complete machine-readable audit is `data/frozen/data_audit.json`; the clean-room receipt is `data/frozen/clean_room_receipt.json`.